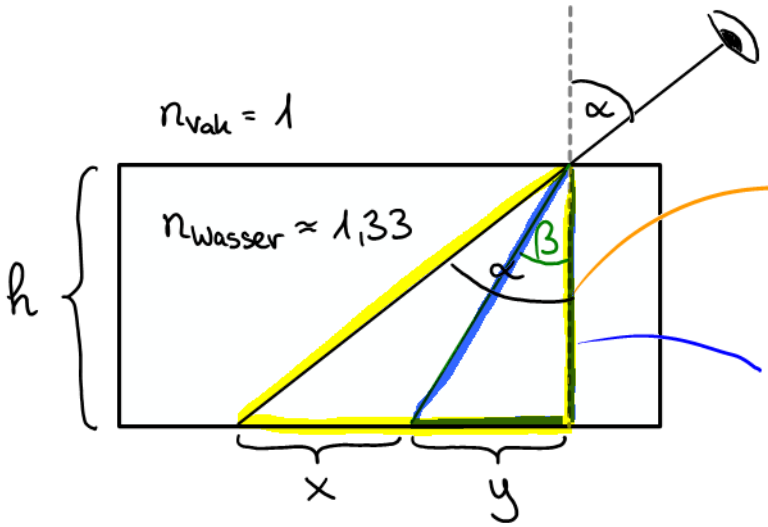
Musterlösung - Lichtbrechung einer Münze



$$\tan(\alpha) = \frac{x+y}{h}$$

$$\tan(\beta) = \frac{y}{h}$$

Berechne Winkel β: $\quad n_{vak} \cdot \sin(\alpha) = n_w \cdot \sin(\beta)$

$$\sin(\beta) = \sin(\alpha)\ \frac{n_{vak}}{n_w}$$

$$\beta = \arcsin\left(\sin(\alpha)\ \frac{n_{vak}}{n_w}\right)$$

$$\approx \arcsin\left(\sin(45°)\ \frac{1}{1{,}33}\right) \approx \underline{32{,}18°}$$

Berechne y: $\quad \tan(\beta) = \frac{y}{h}$

$$y = \tan(\beta) \cdot h = \tan(32{,}18°) \cdot 1m = \underline{0{,}628\,m}$$

Berechne x: $\quad \tan(\alpha) = \frac{x+y}{h}$

$$x = \tan(\alpha) \cdot h - y \approx \tan(45°) \cdot 1m - 0628m \approx \underline{0{,}372m}$$